

JEE Main Level Practice Test-11

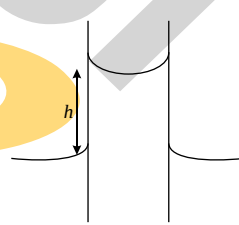
for JEE & NEET Aspirants

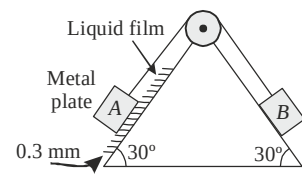
**Topic : SURFACE TENSION,
VISCOSITY & ELASTICITY**

Time: 75Min Marking +4 -1

Section - A : MCQs with Single Option Correct

- Work done in increasing the size of a soap bubble from a radius of 3 cm to 5 cm is nearly (Surface tension of soap solution = 0.03 Nm^{-1}):
 (A) $2\pi \text{ mJ}$ (B) $0.4\pi \text{ mJ}$ (C) $4\pi \text{ mJ}$ (D) $0.2\pi \text{ mJ}$
- Assume that a drop of liquid evaporates by decrease in its surface energy, so that its temperature remains unchanged. What should be the minimum radius of the drop for this to be possible? The surface tension is T , density of liquid is ρ and L is its latent heat of vaporization :
 (A) $\sqrt{T/\rho L}$ (B) $T/\rho L$ (C) $2T/\rho L$ (D) $\rho L/T$
- In a capillary rise experiment, radius of curvature of liquid surface (approximately spherical) is R . If approximate rise in h , the surface tension of the liquid is : (density of liquid is ρ)
 (A) $\frac{\rho ghR}{4}$ (B) $\frac{\rho ghR}{2}$ (C) Zero (D) ρghR
- A sphere of mass m and radius r is projected in a gravity free space with speed v . If coefficient of viscosity is $\frac{1}{6\pi}$, the distance traveled by the body before it stops is : (Consider Stoke's law to be valid during retardation)
 (A) $\frac{mv}{2r}$ (B) $\frac{2mv}{r}$ (C) $\frac{mv}{r}$ (D) None of these


- A metal plate of area 0.10 m^2 is connected to a 0.04 kg mass via a string that passes over an ideal pulley (considered massless and frictionless) A liquid with a film of thickness of 0.3 mm is placed between the plate and the table. When system is released, the plate moves up with constant speed of 0.085 m/s . Find the approximate value of coefficient of viscosity of the liquid. (Mass of metal plate is 0.02 kg , $g = 9.8 \text{ m/s}^2$):
 (A) $3.46 \times 10^{-3} \text{ Pa sec}$
 (B) $3.46 \times 10^{-4} \text{ Pa sec}$
 (C) $6.90 \times 10^{-3} \text{ Pa sec}$
 (D) $6.90 \times 10^{-4} \text{ Pa sec}$


- Two solid spherical balls of radius r_1 & r_2 ($< r_1$) & of density σ are tied up with a long string & released in a viscous liquid column of lesser density ρ with the string just taut initially. Find the tension in the string when terminal velocity is attained :
 (A) $\frac{4}{3}\pi\left(\frac{r_2^4 - r_1^4}{r_2 - r_1}\right)(\sigma - \rho)g$ (B) $\frac{2}{3}\pi(r_2^3 - r_1^3)(\rho - g)g$

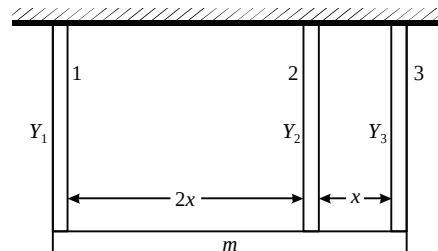
(C) $\frac{4}{3}\pi(r_2^3 - r_1^3)(\sigma - \rho)g$

(D) $\frac{4}{3}\pi\left(\frac{r_2^4 - r_1^4}{r_2 + r_1}\right)(\sigma - \rho)g$

7. A rain drop of radius 1.5 mm, experiences a drag force $F = (2 \times 10^{-5}v)$ N, while falling through air from a height 2 km, with a velocity v . The terminal velocity of the rain drop will be nearly (use $g = 10 \text{ m/s}^2$):
 (A) 200 m/s (B) 80 m/s (C) 4.7 m/s (D) 3 m/s
8. A spherical object of mass 1 kg and radius 1 m is falling vertically downward inside a viscous liquid in a gravity free space. At a certain instant the velocity of the sphere is 2 m/s. If the coefficient of viscosity of the liquid is $\frac{1}{18\pi} \text{ N-s/m}^2$, then velocity of ball will become 0.5 m/s after a time :
 (A) $\ln 4 \text{ s}$ (B) $2 \ln 4 \text{ s}$ (C) $3 \ln 4 \text{ s}$ (D) $2 \ln 2 \text{ s}$
9. A viscous liquid flows through a uniform tube. As we move from the axis of the tube towards its wall, the speed of the liquid relative to the tube :
 (A) Increases from zero to maximum value
 (B) Decreases from maximum value to zero
 (C) Remains same throughout the cross section
 (D) Increases from zero to a maximum value on one side and decreases from a maximum value to zero on the other side.
10. If two uniform spherical drops of a liquid having same mass, radius, and falling freely with a terminal velocity of $\sqrt[3]{16} \text{ m/s}$, coalesce to form a bigger drop, then the terminal velocity with which the bigger drop falls is :
 (A) 8 m/s (B) 4 m/s (C) 10 m/s (D) None of these
11. A uniform wire (Young's modulus $2 \times 10^{11} \text{ Nm}^{-2}$) is subjected to longitudinal tensile stress of $5 \times 10^7 \text{ Nm}^{-2}$. If the overall volume change in the wire is 0.02%, the fractional decrease in the radius of the wire is close to :
 (A) 1.0×10^{-4} (B) 1.5×10^{-4} (C) 0.25×10^{-4} (D) 5×10^{-4}
12. A copper wire of length 1.0 m and a steel wire of length 0.5 m having equal cross-sectional areas are joined end to end. The composite wire is stretched by a certain load which stretches the copper wire by 1 mm. If Young's moduli of copper and steel are respectively $1.0 \times 10^{11} \text{ Nm}^{-2}$ and $2.0 \times 10^{11} \text{ Nm}^{-2}$, the total extension of the composite wire is :
 (A) 1.75 mm (B) 2.0 mm (C) 1.50 mm (D) 1.25 mm
13. In materials like aluminium and copper, the correct order of magnitude of various elastic moduli is :
 (A) Young's modulus < shear modulus < bulk modulus
 (B) Bulk modulus < shear modulus < Young's modulus
 (C) Shear modulus < Young's modulus < bulk modulus
 (D) Bulk modulus < Young's modulus < shear modulus
14. The bulk moduli of ethanol, mercury and water are given as 0.9, 25 and 2.2 respectively in units of 10^9 Nm^{-2} . For a given value of pressure, the fractional compression in volume is $\frac{\Delta V}{V}$. Which of the following statements about $\frac{\Delta V}{V}$ for these three liquids is correct ?
 (A) Ethanol > Water > Mercury (B) Water > Ethanol > Mercury
 (C) Mercury > Ethanol > Water (D) Ethanol > Mercury > Water

15. Steel ruptures when a shear of $3.5 \times 10^8 \text{ Nm}^{-2}$ is applied. The force needed to punch a 1 cm diameter hole in a steel sheet 0.3 cm thick is nearly :
 (A) $1.4 \times 10^4 \text{ N}$ (B) $2.7 \times 10^4 \text{ N}$ (C) $3.3 \times 10^4 \text{ N}$ (D) $1.1 \times 10^4 \text{ N}$

16. Three vertical wires 1, 2 and 3 with young's modulus Y_1 , Y_2 and Y_3 respectively are supporting a block of mass m in horizontal position. The wires are of equal length and cross section area. It is given that $Y_2 = Y_3$. The wires 1 and 3 are attached at extreme ends of the block. The tension in the wires are related as :

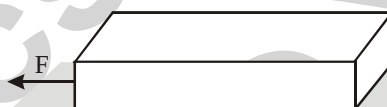


- (A) $T_1 = T_2/2$
 (B) $2T_1 = T_3$
 (C) $T_2 = T_3$
 (D) $T_1 = \frac{mg}{5}$

17. In the Searle's experiment of determination of Young's modulus of a wire, the quantity which should be measured with greatest precision is :

- (A) Diameter of the wire
 (B) Length of the wire
 (C) Deflection of loaded wire
 (D) Loading weights

18. A uniform bar of square cross-section is lying along a frictionless horizontal surface. A horizontal force is applied to pull it from one of its ends then:



- (A) The bar is under same stress throughout its length
 (B) The bar is not under any stress because force has been applied only at one end
 (C) The bar simply moves without any stress in it
 (D) The stress developed reduces to zero at the end of the bar where no force is applied

19. A metal string is fixed between two rigid supports. Initially, a negligible tension exists in the string. If the string has density d , young's modulus Y and coefficient of thermal expansion α and is allowed to cool through a temperature t then the transverse waves will move along it with speed :

- (A) $t\sqrt{\frac{Y\alpha}{d}}$ (B) $\sqrt{\frac{Y\alpha t}{d}}$ (C) $\alpha t\sqrt{\frac{Y}{d}}$ (D) $Y\sqrt{\frac{\alpha t}{d}}$

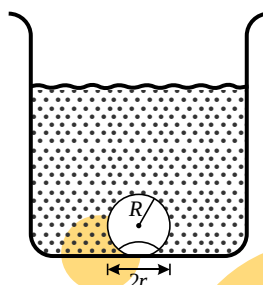
20. Water has a bulk modulus $2.2 \times 10^9 \text{ Pa}$ when a litre of water is put under a pressure of 2 atm, what is the increase in elastic potential energy stored in the water ? [1 atm = 10^5 Pa]
 (A) $9.1 \times 10^{-3} \text{ J}$ (B) $4.5 \times 10^{-3} \text{ J}$ (C) $2.3 \times 10^{-3} \text{ J}$ (D) $6.8 \times 10^{-3} \text{ J}$

Section- B: INTEGER Answer Type Questions

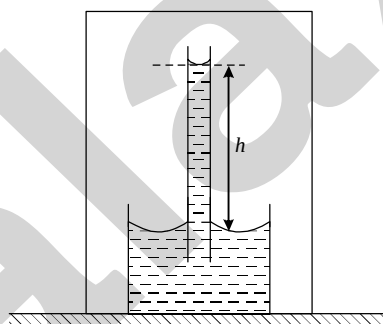
21. A 20 cm long capillary tube is dipped in water. The water rises upto 8 cm. If the entire arrangement is put in a freely falling elevator, calculate the length of water column in the capillary tube.

22. A large number of liquid drops each of radius r coalesce to form a single drop of radius R . The energy released in the process is converted into kinetic energy of the big drop so formed. The speed of the big drop is (given, surface tension of liquid T , density ρ) $\sqrt{\frac{nT}{\rho} \left(\frac{1}{r} - \frac{1}{R} \right)}$ then calculate n .

23. On heating water of density ρ_w , bubbles being formed at the bottom of the vessel detach and rise. Take the bubbles to be spheres of radius R and making a circular contact of radius r with the bottom of the vessel. If $r \ll R$ and the surface tension of water is T , value of r just before bubbles detach is $R^2 \sqrt{\frac{n\rho_w g}{3T}}$ then calculate n .



24. A capillary tube of inner radius $r = 5$ mm is covered in a container with a liquid whose vapour pressure is negligible. The density of the liquid is $\rho = 1000 \text{ kg/m}^3$. The container and the tube are in vacuum condition. Calculate the pressure (N/m^2) inside liquid in the capillary tube at a height $h = 20$ cm as shown in figure. [$g = 10 \text{ m/s}^2$]

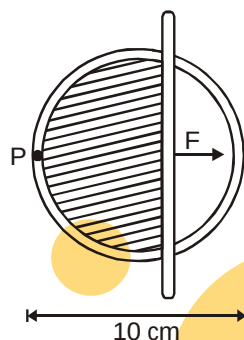


25. A spherical balloon has interior pressure, P_1 and radius r and is in equilibrium inside an enclosure with pressure $P_0 = \frac{8P_1}{9}$. The enclosure is gradually evacuated. Assuming that temperature is fixed and tension per unit length of the balloon material remains constant. The maximum radius of balloon will be $r_{\text{max}} = nr$. Calculate the value of n .
26. Two soap bubbles of radii 2 mm and 4 mm are brought in contact. If the surface tension of liquid is $7 \times 10^{-2} \text{ Nm}^{-1}$. Then calculate the radius (mm) of the common surface.
27. An isolated and charged spherical soap bubble has a radius ' r ' and the pressure inside is atmospheric. If T is the surface tension of soap solution, then charge on drop is $N\pi r \sqrt{2rT\epsilon_0}$, then find the value of N .
28. Two soap bubbles A and B are kept in a closed chamber where the air is maintained at pressure 8 Nm^{-2} . The radii of

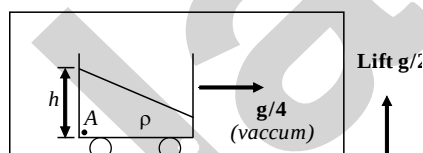
bubbles A and B are 2 cm and 4 cm respectively. Surface tension of the soap-water used to make bubbles is 0.04 Nm^{-1} .

Find the ratio $\frac{n_B}{n_A}$, when n_A and n_B are the number of moles of air in bubbles A and B , respectively. Neglect the effect of gravity and assume temperature of A and B are same.

29. A circular wire, 10 cm in diameter, with a slider wire on it, is in a horizontal plane. A liquid film is formed, bounded by the wires, on the left side of the slider, as shown. The surface tension of the liquid is $100 \times 10^{-3} \text{ N/m}$. An applied force F , perpendicular to the slider, maintains the film in equilibrium. Ignore the sag in the film. In figure, when the slider is 9 cm from point P , the applied force is $x \times 10^{-3} \text{ N}$. Calculate the value of $\frac{x}{6}$.



30. The evacuated lift is accelerating upwards with acceleration of $g/2$. The tank filled with liquid of density ρ is accelerating with $g/4$ with respect to the lift as shown in figure. The pressure at point A at bottom of tank is k times $\frac{3}{2} h \rho g$. Find k .



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ANSWER KEY

Section - A : MCQs with Single Option Correct

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|---------|---------|---------|---------|
| 1. (B) | 2. (C) | 3. (B) | 4. (C) |
| 5. (A) | 6. (D) | 7. (C) | 8. (C) |
| 9. (B) | 10. (B) | 11. (C) | 12. (D) |
| 13. (C) | 14. (A) | 15. (C) | 16. (C) |
| 17. (A) | 18. (D) | 19. (B) | 20. (D) |

Section- B: INTEGER Answer Type Questions

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| 21. [0] | 22. [6] | 23. [2] | 24. [2000] |
| 25. [3] | 26. [4] | 27. [8] | 28. [6] |
| 29. [2] | 30. [1] | | |

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